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(54) **TITANIUM DIOXIDE CATALYST
STRUCTURE FOR PROCESSES UP TO 1000°
C AND MANUFACTURING THEREOF**

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(52) **U.S. Cl.**
USPC **502/208; 502/350; 423/610; 977/773**

(58) **Field of Classification Search** 423/610;
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See application file for complete search history.

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(57) **ABSTRACT**

The TiO₂ catalyst structure consisting of TiO₂ nano-particles
in the anatase crystal form, doped with 0.05-5 wt % phospho-
rus on the TiO₂ basis, organized in the circular planar aggre-
gates with the specific surface area ranging from 40 to 120
m²/g, suitable for catalytic processes at the temperature up to
800° C., and the TiO₂ catalyst structure of with the morphol-
ogy of the aggregated compact particles, with the specific
surface area from 20 to 40 m²/g, suitable for the catalytic
processes at the temperature up to 1000° C. Active substances
selected from the group consisting of silver, copper, gold,
platinum metals, nickel, molybdenum and metal oxides
except for alkaline metals oxides can be applied onto the
surface of both types of the structure.

12 Claims, 5 Drawing Sheets

